Simulating Auto Claims Amounts Jarrett Baker

Jarrett Baker

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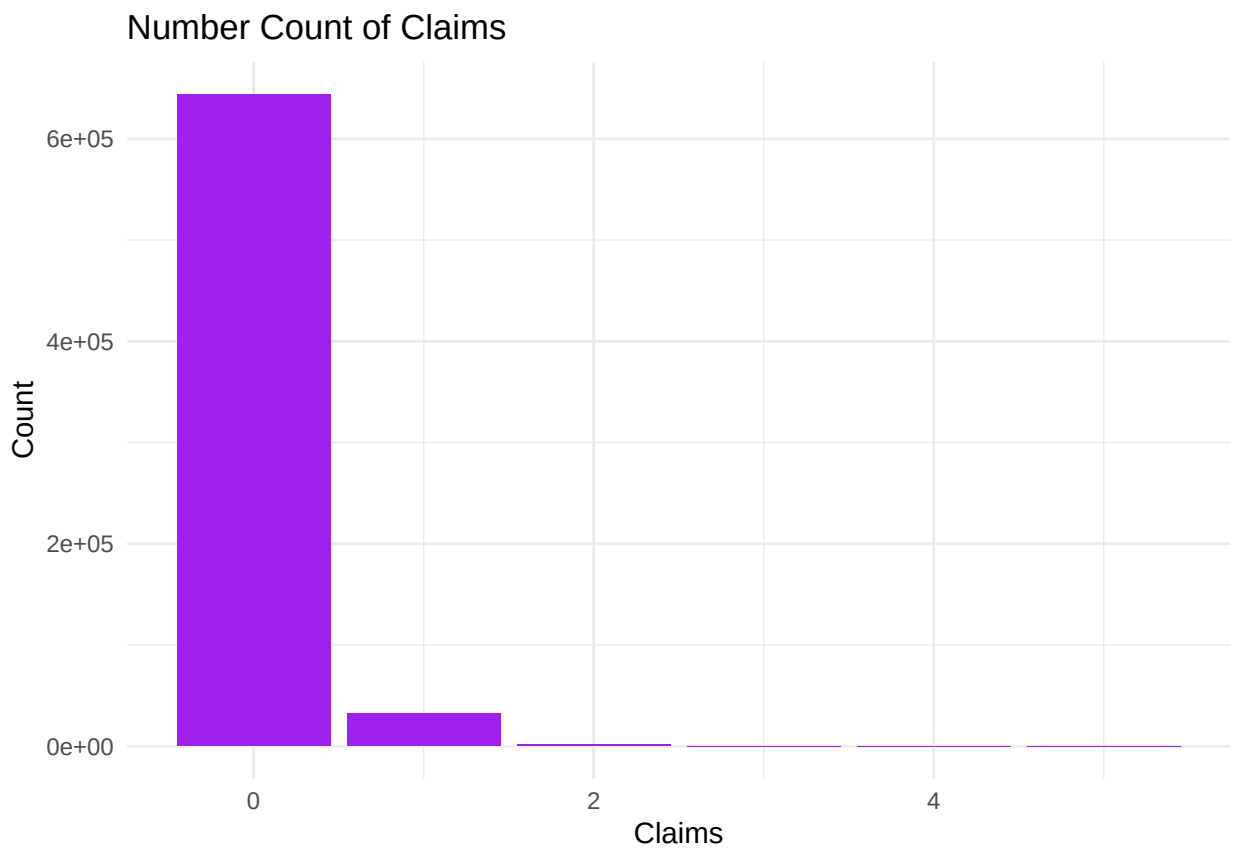
```r
library(tidyverse)
library(MASS)          # NB GLM
library(broom)         # Tidy model outputs
library(performance)   # Dispersion checks & diagnostics
library(pscl)          # ZINB
library(ggplot2)       # Creating visuals for variables
library(lme4)
```

```r
french_data <- read.csv("~/freMTPL2freq.csv")

summary(french_data)
```

```
##      IDpol            ClaimNb            Exposure           VehPower
##  Min.   :      1   Min.   : 0.00000   Min.   :0.002732   Min.   : 4.000
##  1st Qu.:1157951   1st Qu.: 0.00000   1st Qu.:0.180000   1st Qu.: 5.000
##  Median :2272152   Median : 0.00000   Median :0.490000   Median : 6.000
##  Mean   :2621857   Mean   : 0.05325   Mean   :0.528750   Mean   : 6.455
##  3rd Qu.:4046274   3rd Qu.: 0.00000   3rd Qu.:0.990000   3rd Qu.: 7.000
##  Max.   :6114330   Max.   :16.00000   Max.   :2.010000   Max.   :15.000
##      VehAge           DrivAge        BonusMalus       VehBrand
##  Min.   :  0.000   Min.   : 18.0   Min.   : 50.00   Length:678013
##  1st Qu.:  2.000   1st Qu.: 34.0   1st Qu.: 50.00   Class :character
##  Median :  6.000   Median : 44.0   Median : 50.00   Mode  :character
##  Mean   :  7.044   Mean   : 45.5   Mean   : 59.76
##  3rd Qu.: 11.000   3rd Qu.: 55.0   3rd Qu.: 64.00
##  Max.   :100.000   Max.   :100.0   Max.   :230.00
##     VehGas              Area              Density         Region
##  Length:678013      Length:678013      Min.   :    1   Length:678013
##  Class :character   Class :character   1st Qu.:   92   Class :character
##  Mode  :character   Mode  :character   Median :  393   Mode  :character
##                                        Mean   : 1792
##                                        3rd Qu.: 1658
##                                        Max.   :27000
```

```r
#Claim Number on historical data
ggplot(french_data, aes(x = pmin(ClaimNb, 5))) +
  geom_bar(fill = "purple") +
  theme_minimal() +
  labs(title = "Number Count of Claims", x = "Claims", y = "Count")
```

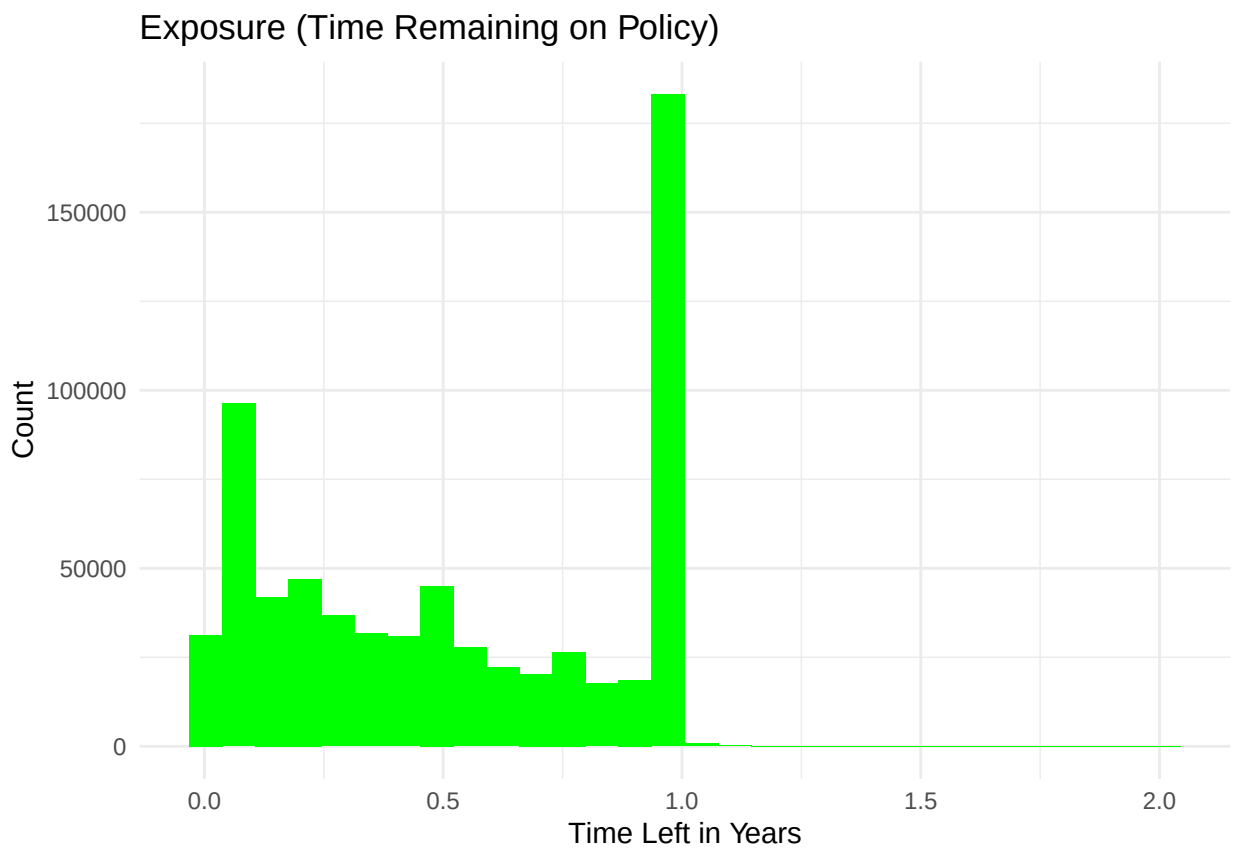



```
french_data %>% count(ClaimNb)
```

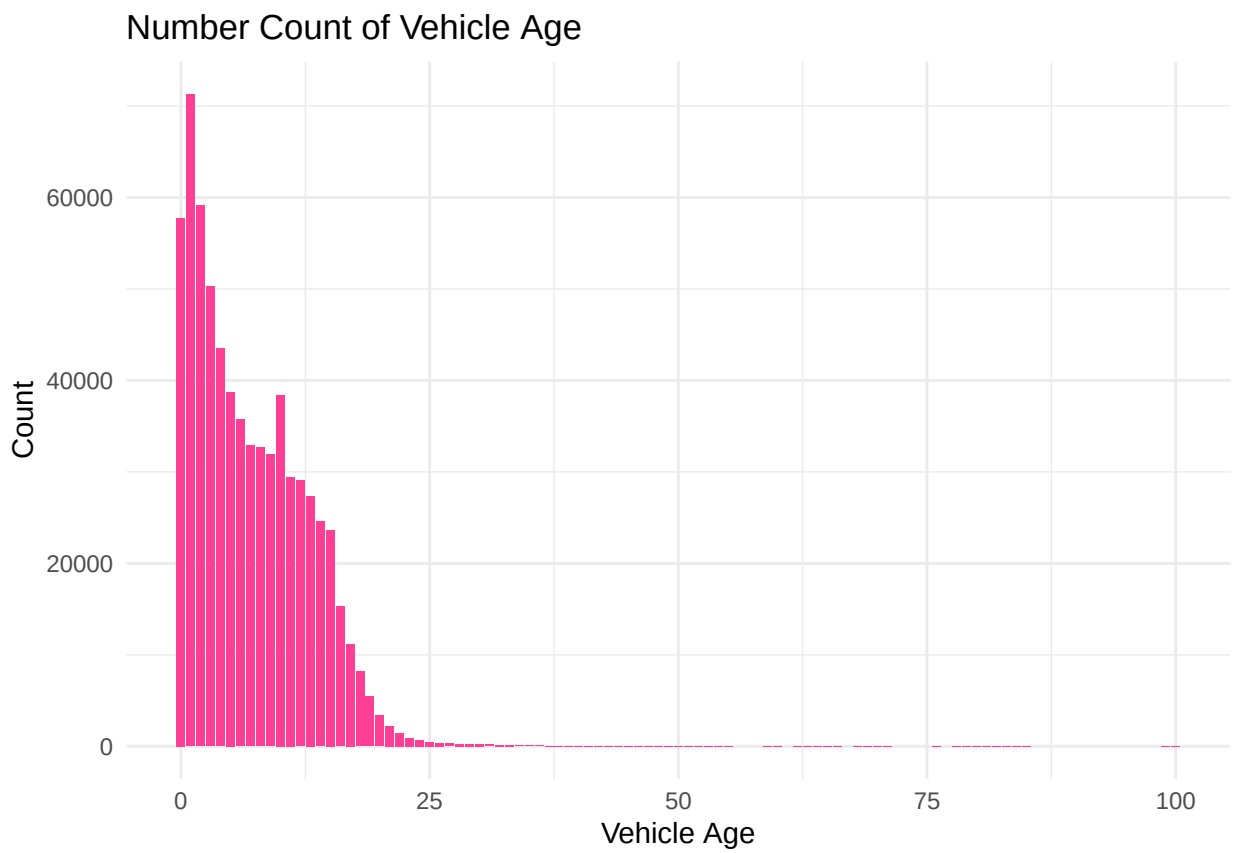
```
##    ClaimNb      n
## 1        0 643953
## 2        1  32178
## 3        2   1784
## 4        3     82
## 5        4      7
## 6        5      2
## 7        6      1
## 8        8      1
## 9        9      1
## 10      11      3
## 11      16      1
```

```
# Exposure
ggplot(french_data, aes(x = Exposure)) +
  geom_histogram(fill = "green") +
  theme_minimal() +
  labs(title = "Exposure (Time Remaining on Policy)", x = "Time Left in Years", y = "Count")
```

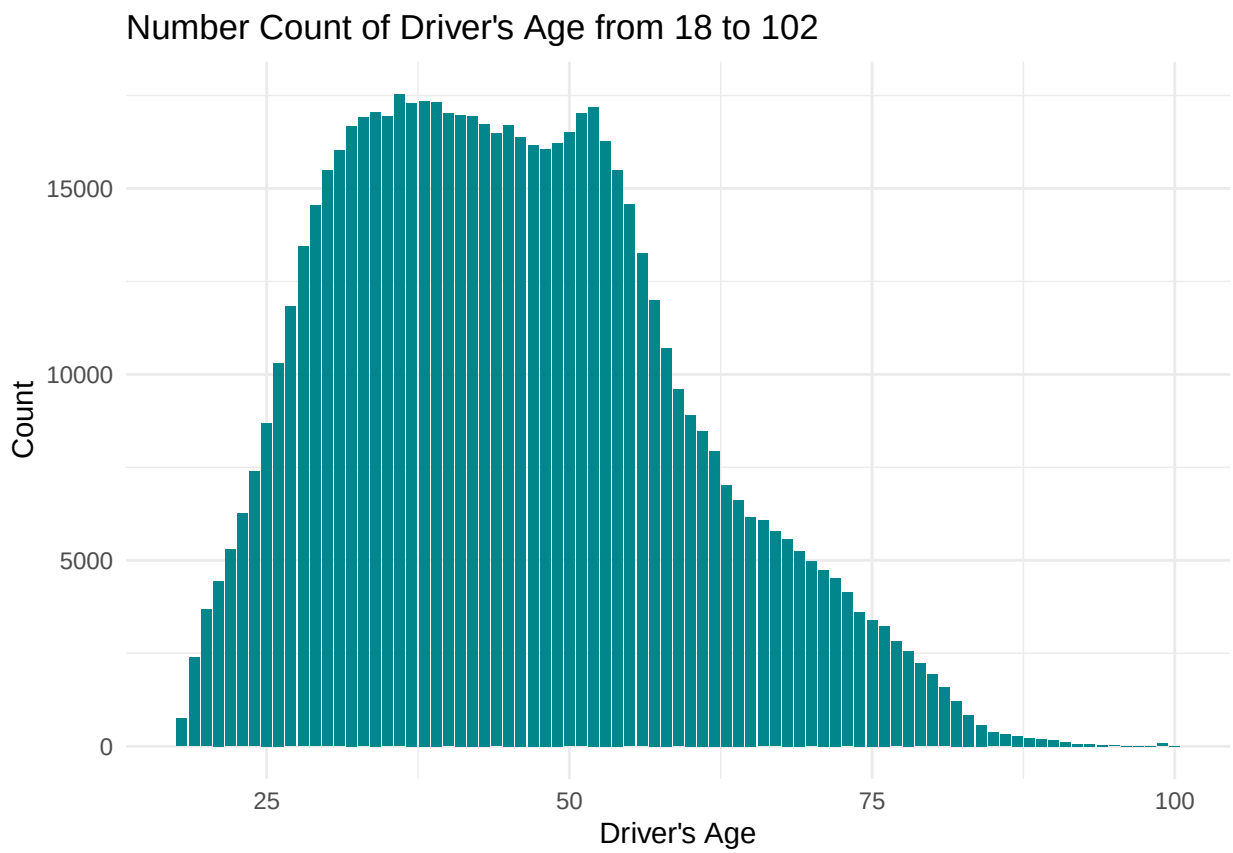
```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```r
# Vehicle Age
ggplot(french_data, aes(x = VehAge)) +
  geom_bar(fill = "violetred1") +
  theme_minimal() +
  labs(title = "Number Count of Vehicle Age", x = "Vehicle Age", y = "Count")
```

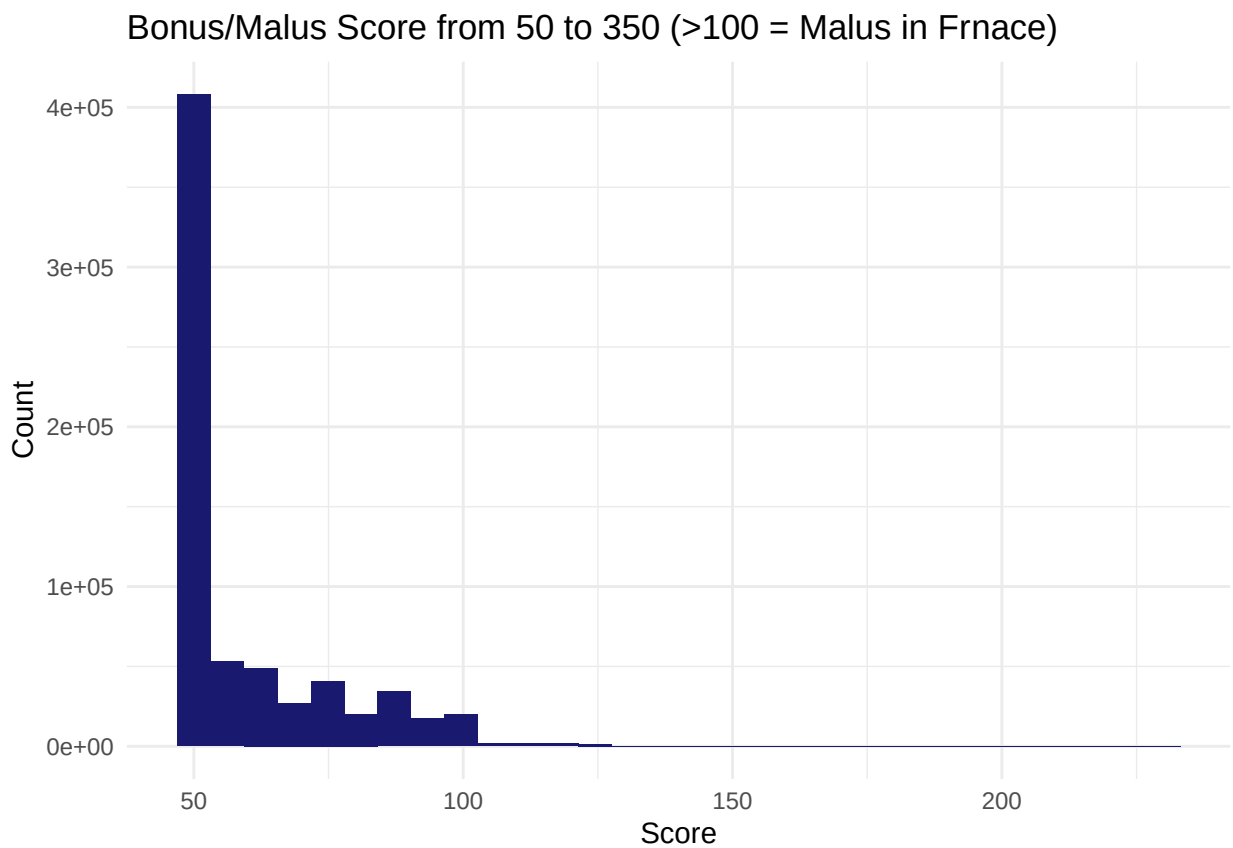



```r
# Driver's Age
ggplot(french_data, aes(x = DrivAge)) +
  geom_bar(fill = "turquoise4") +
  theme_minimal() +
  labs(title = "Number Count of Driver's Age from 18 to 102", x = "Driver's Age", y = "Count")
```

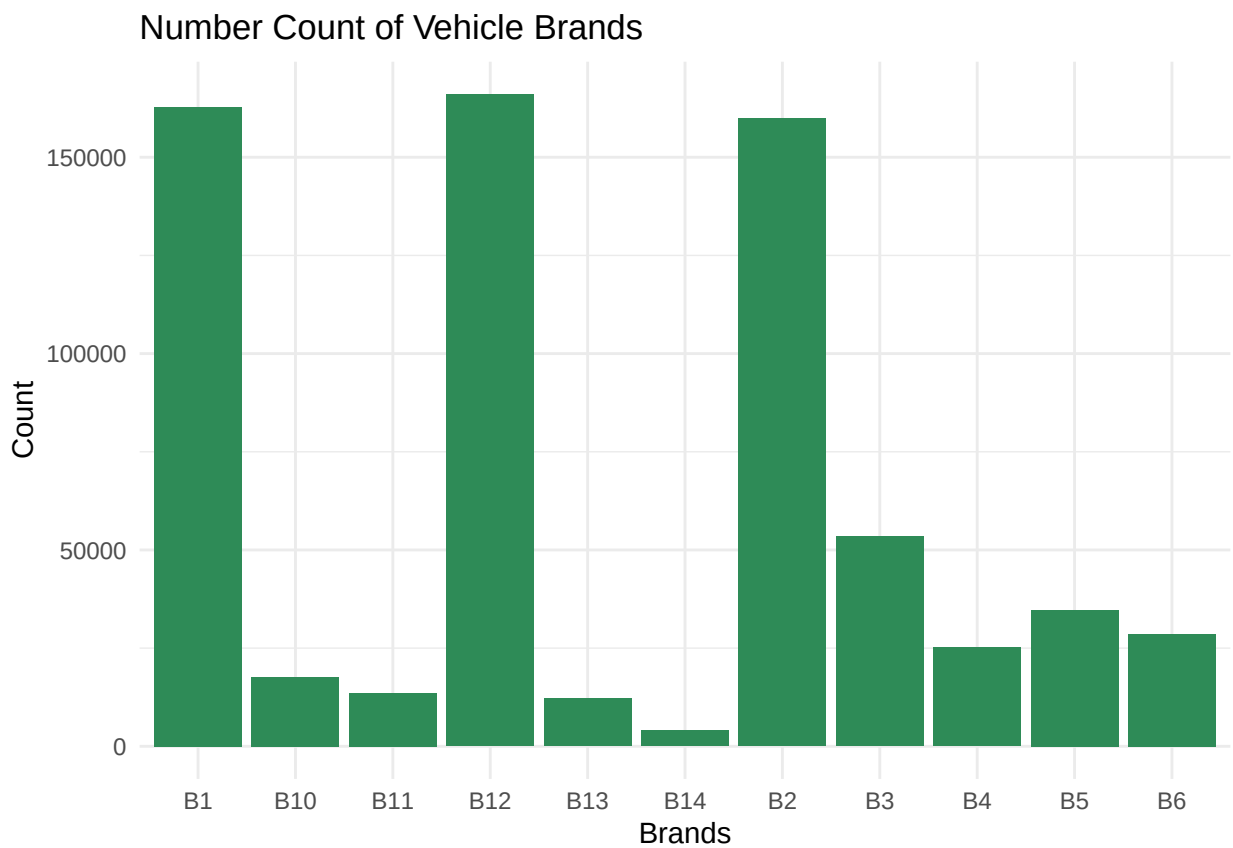



```r
# Bonus/Malus Score
ggplot(french_data, aes(x = BonusMalus)) +
  geom_histogram(fill = "midnightblue") +
  theme_minimal() +
  labs(title = "Bonus/Malus Score from 50 to 350 (>100 = Malus in Frnace)", x = "Score", y = "Count")
```

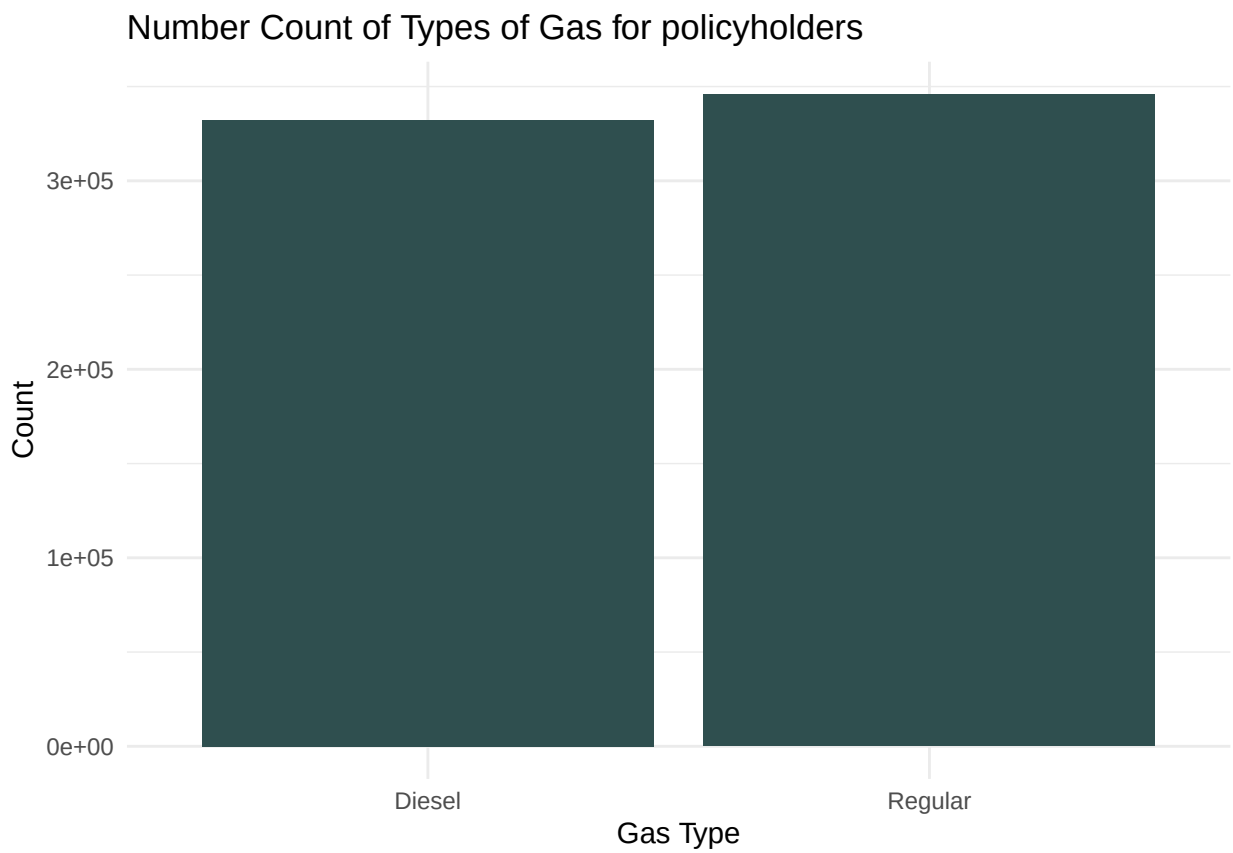
```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



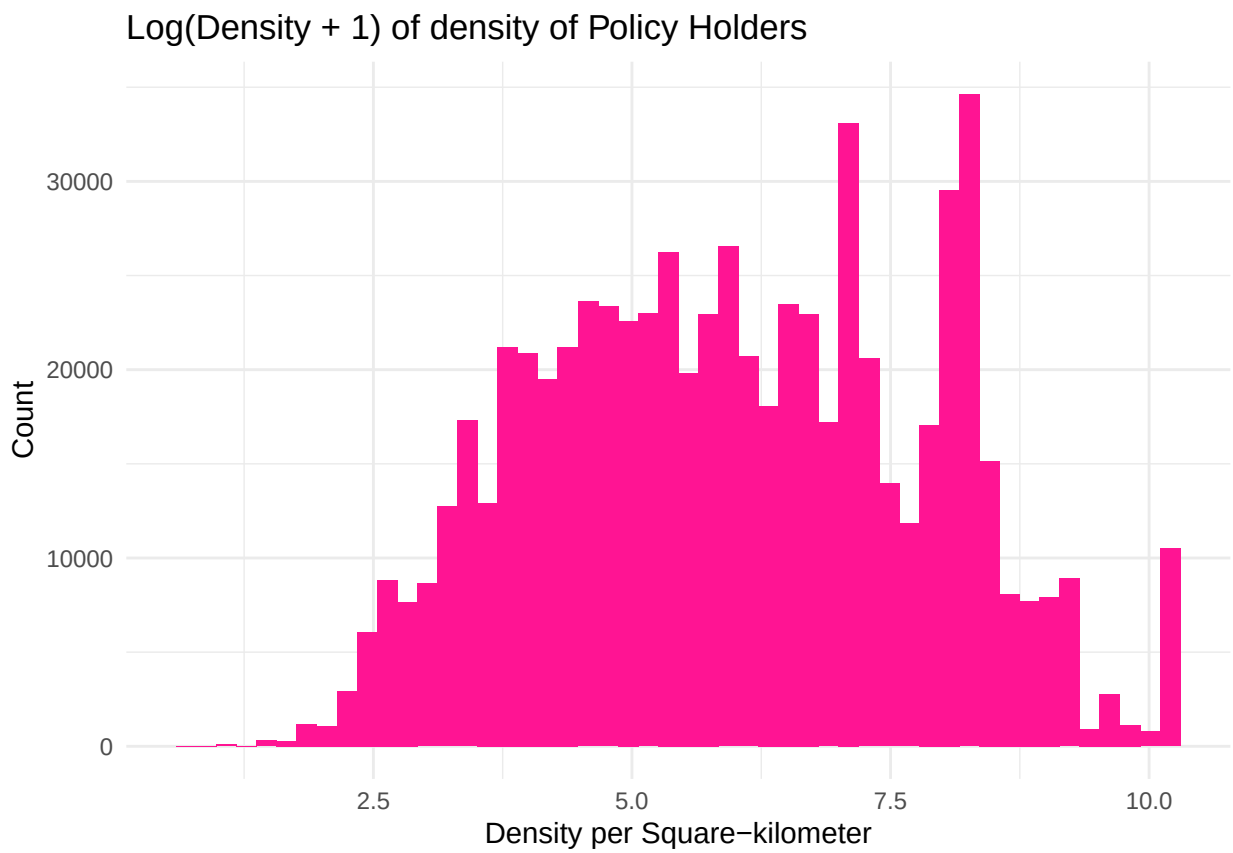

```r
# Vehicle Brand
ggplot(french_data, aes(x = VehBrand)) +
  geom_bar(fill = "seagreen") +
  theme_minimal() +
  labs(title = "Number Count of Vehicle Brands", x = "Brands", y = "Count")
```



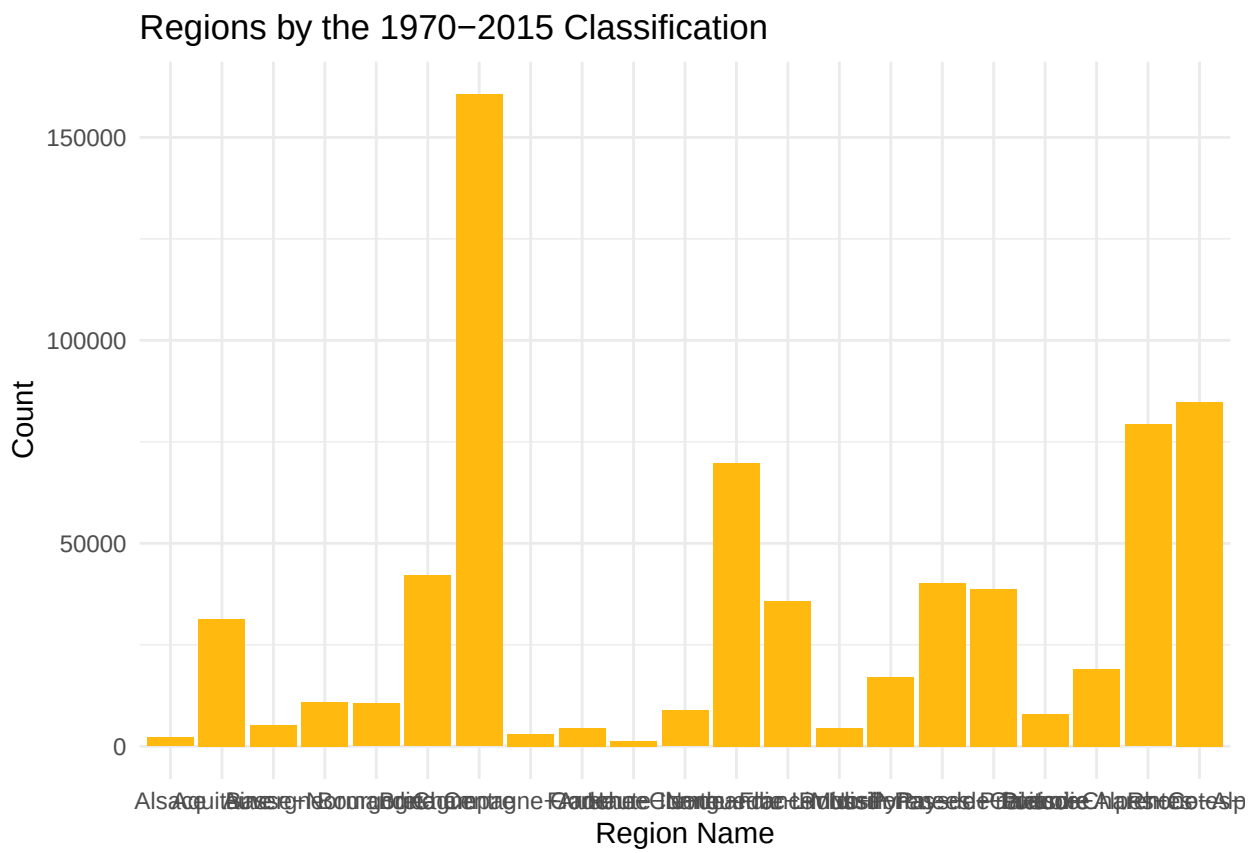

```r
# Vehicle Gas Type
ggplot(french_data, aes(x = VehGas)) +
  geom_bar(fill = "darkslategray") +
  theme_minimal() +
  labs(title = "Number Count of Types of Gas for policyholders", x = "Gas Type", y = "Count")
```




```
ggplot(french_data, aes(x = log(Density + 1))) +
  geom_histogram(bins = 50, fill = "deeppink1") +
  theme_minimal() +
  labs(title = "Log(Density + 1) of density of Policy Holders", x = "Density per Square-kilometer", y =
```




```
ggplot(french_data, aes(x = Region)) +
  geom_bar(fill = "darkgoldenrod1") +
  theme_minimal() +
  labs(title = "Regions by the 1970-2015 Classification", x = "Region Name", y = "Count")
```




```r
french_clean <- french_data %>%
  filter(Exposure > 0) %>%
  mutate(
    Region = as.factor(Region),
    VehBrand = as.factor(VehBrand),
    VehPower = as.factor(VehPower),
    VehGas = as.factor(VehGas),
    VehAge = as.numeric(VehAge),
    DrivAge = as.numeric(DrivAge)
    )

summary(french_clean)
```

```
##      IDpol            ClaimNb            Exposure           VehPower
##  Min.   :      1   Min.   : 0.00000   Min.   :0.002732   6      :148976
##  1st Qu.:1157951   1st Qu.: 0.00000   1st Qu.:0.180000   7      :145401
##  Median :2272152   Median : 0.00000   Median :0.490000   5      :124821
##  Mean   :2621857   Mean   : 0.05325   Mean   :0.528750   4      :115349
##  3rd Qu.:4046274   3rd Qu.: 0.00000   3rd Qu.:0.990000   8      : 46956
##  Max.   :6114330   Max.   :16.00000   Max.   :2.010000   10     : 31354
##                                                          (Other): 65156
##      VehAge           DrivAge        BonusMalus        VehBrand
##  Min.   :  0.000   Min.   : 18.0   Min.   : 50.00   B12    :166024
##  1st Qu.:  2.000   1st Qu.: 34.0   1st Qu.: 50.00   B1     :162736
##  Median :  6.000   Median : 44.0   Median : 50.00   B2     :159861
```

```
##  Mean   :  7.044   Mean   : 45.5   Mean   : 59.76   B3     : 53395
##  3rd Qu.: 11.000   3rd Qu.: 55.0   3rd Qu.: 64.00   B5     : 34753
##  Max.   :100.000   Max.   :100.0   Max.   :230.00   B6     : 28548
##                                                     (Other): 72696
##      VehGas           Area              Density
##  Diesel :332136   Length:678013      Min.   :    1
##  Regular:345877   Class :character   1st Qu.:   92
##                   Mode  :character   Median :  393
##                                      Mean   : 1792
##                                      3rd Qu.: 1658
##                                      Max.   :27000
##
##                          Region
##  Centre                     :160601
##  Rhone-Alpes                : 84752
##  Provence-Alpes-Cotes-D'Azur: 79315
##  Ile-de-France              : 69791
##  Bretagne                   : 42122
##  Nord-Pas-de-Calais         : 40275
##  (Other)                    :201157
```

```r
#GLM Poisson
french_poisson <- glm(
  ClaimNb ~ VehPower + VehAge + DrivAge + BonusMalus + Density + Region,
  offset = log(Exposure),
  family = poisson(link = "log"),
  data = french_clean
)

summary(french_poisson)
```

```
##
## Call:
## glm(formula = ClaimNb ~ VehPower + VehAge + DrivAge + BonusMalus +
##     Density + Region, family = poisson(link = "log"), data = french_clean,
##     offset = log(Exposure))
##
## Coefficients:
##                                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)                      -3.791e+00  9.359e-02 -40.505  < 2e-16 ***
## VehPower5                         1.109e-01  1.773e-02   6.256 3.94e-10 ***
## VehPower6                         1.171e-01  1.724e-02   6.791 1.11e-11 ***
## VehPower7                         8.501e-02  1.756e-02   4.840 1.30e-06 ***
## VehPower8                        -6.839e-02  2.642e-02  -2.588  0.00965 **
## VehPower9                         2.817e-01  2.737e-02  10.294  < 2e-16 ***
## VehPower10                        2.533e-01  2.718e-02   9.321  < 2e-16 ***
## VehPower11                        1.712e-01  3.600e-02   4.754 1.99e-06 ***
## VehPower12                        1.787e-02  5.450e-02   0.328  0.74298
## VehPower13                        1.133e-01  8.198e-02   1.383  0.16681
## VehPower14                        1.710e-01  9.267e-02   1.846  0.06494 .
## VehPower15                       -4.631e-02  9.090e-02  -0.509  0.61044
## VehAge                           -4.320e-02  1.080e-03 -40.015  < 2e-16 ***
## DrivAge                           6.922e-03  3.887e-04  17.808  < 2e-16 ***
## BonusMalus                        2.297e-02  3.041e-04  75.549  < 2e-16 ***
```

```
## Density                            6.140e-06  1.476e-06   4.160 3.19e-05 ***
## RegionAquitaine                   -1.012e-01  9.090e-02  -1.114  0.26549
## RegionAuvergne                    -2.908e-01  1.132e-01  -2.569  0.01020 *
## RegionBasse-Normandie             -4.307e-02  9.541e-02  -0.451  0.65171
## RegionBourgogne                   -6.299e-02  9.737e-02  -0.647  0.51770
## RegionBretagne                     1.991e-02  8.886e-02   0.224  0.82269
## RegionCentre                      -2.896e-02  8.741e-02  -0.331  0.74038
## RegionChampagne-Ardenne            1.371e-01  1.174e-01   1.168  0.24294
## RegionCorse                        1.350e-01  1.077e-01   1.254  0.20999
## RegionFranche-Comte               -1.185e-01  1.583e-01  -0.748  0.45421
## RegionHaute-Normandie             -8.046e-02  1.040e-01  -0.773  0.43925
## RegionIle-de-France                4.136e-02  8.884e-02   0.466  0.64154
## RegionLanguedoc-Roussillon        -1.920e-02  9.039e-02  -0.212  0.83176
## RegionLimousin                     1.244e-01  1.067e-01   1.166  0.24361
## RegionMidi-Pyrenees               -1.118e-01  9.475e-02  -1.180  0.23782
## RegionNord-Pas-de-Calais          -1.479e-01  8.990e-02  -1.645  0.10002
## RegionPays-de-la-Loire            -4.515e-02  8.957e-02  -0.504  0.61425
## RegionPicardie                     6.072e-02  9.901e-02   0.613  0.53971
## RegionPoitou-Charentes            -1.034e-01  9.253e-02  -1.117  0.26397
## RegionProvence-Alpes-Cotes-D'Azur  2.549e-02  8.819e-02   0.289  0.77260
## RegionRhone-Alpes                  6.761e-02  8.787e-02   0.769  0.44161
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 224481  on 678012  degrees of freedom
## Residual deviance: 217441  on 677977  degrees of freedom
## AIC: 286836
##
## Number of Fisher Scoring iterations: 6
```

```r
# Checking for overdispersion (variance / mean)
check_overdispersion(french_poisson)
```

```
## # Overdispersion test
##
##        dispersion ratio =       2.716
##   Pearson's Chi-Squared = 1841084.358
##                 p-value =     < 0.001
```

```
## Overdispersion detected.
```

```r
# Significant overdispersion is present, ratio = 2.7186
# >1 signifies overdispersion (up to ~1.3 is acceptable), >_2 means significant overdispersion
```

```r
#NB as alternate
french_nb <- glm.nb(
  ClaimNb ~ VehPower + VehAge + DrivAge + BonusMalus + Density + Region +
    offset(log(Exposure)),
  data = french_clean
)
```

```r
summary(french_nb)
```

```
##
## Call:
## glm.nb(formula = ClaimNb ~ VehPower + VehAge + DrivAge + BonusMalus +
##     Density + Region + offset(log(Exposure)), data = french_clean,
##     init.theta = 1.057384048, link = log)
##
## Coefficients:
##                                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)                       -3.800e+00  9.773e-02 -38.879  < 2e-16 ***
## VehPower5                          1.111e-01  1.834e-02   6.058 1.38e-09 ***
## VehPower6                          1.144e-01  1.782e-02   6.423 1.34e-10 ***
## VehPower7                          8.276e-02  1.813e-02   4.564 5.03e-06 ***
## VehPower8                         -7.799e-02  2.722e-02  -2.865  0.00417 **
## VehPower9                          2.818e-01  2.837e-02   9.932  < 2e-16 ***
## VehPower10                         2.536e-01  2.814e-02   9.010  < 2e-16 ***
## VehPower11                         1.674e-01  3.719e-02   4.501 6.75e-06 ***
## VehPower12                         1.285e-02  5.607e-02   0.229  0.81868
## VehPower13                         1.054e-01  8.485e-02   1.242  0.21428
## VehPower14                         1.673e-01  9.586e-02   1.745  0.08092 .
## VehPower15                        -4.441e-02  9.314e-02  -0.477  0.63351
## VehAge                            -4.429e-02  1.114e-03 -39.746  < 2e-16 ***
## DrivAge                            7.028e-03  4.071e-04  17.264  < 2e-16 ***
## BonusMalus                         2.330e-02  3.297e-04  70.681  < 2e-16 ***
## Density                            6.385e-06  1.535e-06   4.159 3.19e-05 ***
## RegionAquitaine                   -1.010e-01  9.460e-02  -1.067  0.28582
## RegionAuvergne                    -2.911e-01  1.171e-01  -2.486  0.01293 *
## RegionBasse-Normandie             -4.019e-02  9.931e-02  -0.405  0.68572
## RegionBourgogne                   -5.853e-02  1.013e-01  -0.578  0.56325
## RegionBretagne                     2.308e-02  9.256e-02   0.249  0.80306
## RegionCentre                      -2.812e-02  9.104e-02  -0.309  0.75744
## RegionChampagne-Ardenne            1.525e-01  1.216e-01   1.254  0.20979
## RegionCorse                        1.467e-01  1.117e-01   1.313  0.18917
## RegionFranche-Comte               -1.081e-01  1.631e-01  -0.663  0.50743
## RegionHaute-Normandie             -7.438e-02  1.078e-01  -0.690  0.49040
## RegionIle-de-France                4.979e-02  9.250e-02   0.538  0.59044
## RegionLanguedoc-Roussillon        -8.809e-03  9.405e-02  -0.094  0.92538
## RegionLimousin                     1.338e-01  1.109e-01   1.206  0.22776
## RegionMidi-Pyrenees               -1.055e-01  9.844e-02  -1.072  0.28377
## RegionNord-Pas-de-Calais          -1.451e-01  9.356e-02  -1.551  0.12094
## RegionPays-de-la-Loire            -4.342e-02  9.327e-02  -0.466  0.64153
## RegionPicardie                     6.995e-02  1.030e-01   0.679  0.49692
## RegionPoitou-Charentes            -1.021e-01  9.629e-02  -1.060  0.28898
## RegionProvence-Alpes-Cotes-D'Azur  2.957e-02  9.184e-02   0.322  0.74748
## RegionRhone-Alpes                  7.109e-02  9.152e-02   0.777  0.43732
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(1.0574) family taken to be 1)
##
##     Null deviance: 197243  on 678012  degrees of freedom
## Residual deviance: 190618  on 677977  degrees of freedom
```

```
## AIC: 286016
##
## Number of Fisher Scoring iterations: 1
##
##
##               Theta:  1.0574
##           Std. Err.:  0.0483
##
##  2 x log-likelihood:  -285942.3280
```

```r
# AIC to test which is the better model
AIC(french_nb, french_poisson)
```

```
##                df      AIC
## french_nb      37 286016.3
## french_poisson 36 286836.1
```

```r
# Negative Binomial has the lower AIC
```

```r
# Testing ZINB
# Use DrivAge & BonusMalus because they can influence probabilities of zero claims, other variables aff
french_zinb <- zeroinfl(
  ClaimNb ~ VehPower + VehAge + DrivAge + BonusMalus + VehBrand + VehGas + Region + offset(log(Exposure
            DrivAge + BonusMalus,
  data = french_clean,
  dist = "negbin"
)
summary(french_zinb)
```

```
##
## Call:
## zeroinfl(formula = ClaimNb ~ VehPower + VehAge + DrivAge + BonusMalus +
##     VehBrand + VehGas + Region + offset(log(Exposure)) | DrivAge + BonusMalus,
##     data = french_clean, dist = "negbin")
##
## Pearson residuals:
##      Min       1Q   Median       3Q      Max
##  -0.8826  -0.2652  -0.2036  -0.1133 168.8667
##
## Count model coefficients (negbin with log link):
##                                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)                     -3.0992329  0.1216525 -25.476  < 2e-16 ***
## VehPower5                        0.1570680  0.0189390   8.293  < 2e-16 ***
## VehPower6                        0.1731929  0.0187810   9.222  < 2e-16 ***
## VehPower7                        0.1032124  0.0186075   5.547 2.91e-08 ***
## VehPower8                       -0.0644594  0.0279931  -2.303  0.02130 *
## VehPower9                        0.2729975  0.0288923   9.449  < 2e-16 ***
## VehPower10                       0.2248373  0.0288799   7.785 6.96e-15 ***
## VehPower11                       0.1503345  0.0379407   3.962 7.42e-05 ***
## VehPower12                      -0.0190933  0.0574375  -0.332  0.73957
## VehPower13                       0.0624804  0.0869819   0.718  0.47256
## VehPower14                       0.1625660  0.0969328   1.677  0.09352 .
```

```
## VehPower15                        -0.0928007  0.0942745  -0.984  0.32493
## VehAge                            -0.0401768  0.0012185 -32.972  < 2e-16 ***
## DrivAge                           -0.0026668  0.0008439  -3.160  0.00158 **
## BonusMalus                         0.0190474  0.0005136  37.087  < 2e-16 ***
## VehBrandB10                        0.0409786  0.0385664   1.063  0.28799
## VehBrandB11                        0.1309384  0.0418936   3.125  0.00178 **
## VehBrandB12                        0.1870343  0.0185997  10.056  < 2e-16 ***
## VehBrandB13                        0.0316923  0.0427316   0.742  0.45829
## VehBrandB14                       -0.1269598  0.0810950  -1.566  0.11745
## VehBrandB2                        -0.0105826  0.0159395  -0.664  0.50674
## VehBrandB3                        -0.0039775  0.0229008  -0.174  0.86211
## VehBrandB4                        -0.0199997  0.0310964  -0.643  0.52013
## VehBrandB5                         0.0605979  0.0260649   2.325  0.02008 *
## VehBrandB6                        -0.0204366  0.0296412  -0.689  0.49053
## VehGasRegular                      0.0861976  0.0118839   7.253 4.07e-13 ***
## RegionAquitaine                   -0.0916273  0.0951516  -0.963  0.33557
## RegionAuvergne                    -0.3224712  0.1178257  -2.737  0.00620 **
## RegionBasse-Normandie             -0.0103515  0.0999114  -0.104  0.91748
## RegionBourgogne                   -0.0551700  0.1018747  -0.542  0.58813
## RegionBretagne                     0.0653699  0.0931429   0.702  0.48279
## RegionCentre                       0.0252528  0.0916486   0.276  0.78290
## RegionChampagne-Ardenne            0.1079541  0.1228886   0.878  0.37969
## RegionCorse                        0.0798720  0.1127240   0.709  0.47860
## RegionFranche-Comte               -0.1229602  0.1649931  -0.745  0.45612
## RegionHaute-Normandie             -0.0610085  0.1085901  -0.562  0.57424
## RegionIle-de-France                0.0571604  0.0924148   0.619  0.53623
## RegionLanguedoc-Roussillon        -0.0231534  0.0946301  -0.245  0.80671
## RegionLimousin                     0.1262748  0.1116866   1.131  0.25822
## RegionMidi-Pyrenees               -0.1326944  0.0991206  -1.339  0.18066
## RegionNord-Pas-de-Calais          -0.1382585  0.0941232  -1.469  0.14186
## RegionPays-de-la-Loire            -0.0148968  0.0938338  -0.159  0.87386
## RegionPicardie                     0.0606419  0.1037321   0.585  0.55882
## RegionPoitou-Charentes            -0.0652216  0.0968591  -0.673  0.50071
## RegionProvence-Alpes-Cotes-D'Azur  0.0272686  0.0923801   0.295  0.76786
## RegionRhone-Alpes                  0.1023235  0.0920676   1.111  0.26640
## Log(theta)                         0.2996292  0.0612011   4.896 9.79e-07 ***
##
## Zero-inflation model coefficients (binomial with logit link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  5.962251   0.422283   14.12   <2e-16 ***
## DrivAge     -0.116746   0.008461  -13.80   <2e-16 ***
## BonusMalus  -0.051139   0.003844  -13.30   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Theta = 1.3494
## Number of iterations in BFGS optimization: 89
## Log-likelihood: -1.427e+05 on 50 Df
```

```r
AIC(french_nb, french_zinb)
```

```
##             df      AIC
## french_nb   37 286016.3
## french_zinb 50 285570.8
```

```r
#ZINB Test is better in AIC standards
#Running Vuong Test to ensure that the ZINB test is better
vuong(french_nb, french_zinb)
```

```
## Vuong Non-Nested Hypothesis Test-Statistic:
## (test-statistic is asymptotically distributed N(0,1) under the
##  null that the models are indistinguishible)
## ------------------------------------------------------------
##               Vuong z-statistic             H_A    p-value
## Raw                  -9.880442 model2 > model1 < 2.22e-16
## AIC-corrected        -9.335641 model2 > model1 < 2.22e-16
## BIC-corrected        -6.222944 model2 > model1 2.4396e-10
```

```r
# Zero Inflation Poisson
french_zip <- zeroinfl(ClaimNb ~ VehPower + VehAge + DrivAge + BonusMalus + VehBrand + VehGas + Region +
  data = french_clean,
  dist = "poisson")

AIC(french_zinb, french_zip)
```

```
##             df      AIC
## french_zinb 50 285570.8
## french_zip  49 285808.9
```

```r
# Comparing Different models
# Different functions for different types of models
compute_residual_deviance <- function(y, mu) {
  2 * sum(ifelse(y == 0, mu, y * log(y / mu) - (y - mu)))
}
get_model_stats <- function(model, data, response_var = "ClaimNb") {

  # Predicted mean
  mu <- predict(model, type = "response")
  y <- data[[response_var]]

  # Pearson Chi-square used for relative comparison only
  # Large-sample inflation causes a problem
  pearson <- sum((y - mu)^2 / mu)

  # Actual zero count
  actual_zero_count <- sum(y == 0)

# Predicted zero count
  pred_zero_count <- tryCatch({

  mu <- predict(model, type = "response")
  n  <- length(mu)

  if (inherits(model, "glm")) {
    # Poisson
    sum(exp(-mu))
```

```r
  } else if (inherits(model, "negbin")) {
    # Negative Binomial
    theta <- model$theta
    sum((theta / (theta + mu))^theta)

  } else if (inherits(model, "zeroinfl")) {
    # ZIP / ZINB
    probs <- predict(model, type = "prob")
    sum(probs[, "0"])

  } else {
    NA
  }

}, error = function(e) NA)

  # Residual Deviance (fixed)
  dev <- tryCatch(deviance(model), error = function(e) NA)

  if (length(dev) == 0 || is.na(dev)) {
    # compute Poisson pseudo-deviance
    dev <- compute_residual_deviance(y, mu)
  }

  tibble(
    N = nrow(data),
    DF_model = length(coef(model)),
    DF_resid = df.residual(model),
    AIC = AIC(model),
    LogLik = as.numeric(logLik(model)),
    Deviance = dev,
    Pearson_ChiSq = pearson,
    Predicted_Zero = pred_zero_count,
    Actual_Zero = actual_zero_count
  )
}


poi_stats  <- get_model_stats(french_poisson, french_clean)
nb_stats   <- get_model_stats(french_nb, french_clean)
zip_stats  <- get_model_stats(french_zip, french_clean)
zinb_stats <- get_model_stats(french_zinb, french_clean)

comparison_table <- bind_rows(
  poi_stats,
  nb_stats,
  zip_stats,
  zinb_stats
)

comparison_table$Model <- c("Poisson", "NegBin", "ZIP", "ZINB")
```

```r
comparison_long <- comparison_table %>%
  pivot_longer(
    cols = -Model,
    names_to = "Metric",
    values_to = "Value"
  )

comparison_wide <- comparison_long %>%
  pivot_wider(
    names_from = Model,
    values_from = Value
  )

comparison_wide
```

```
## # A tibble: 9 x 5
##   Metric          Poisson   NegBin      ZIP     ZINB
##   <chr>             <dbl>    <dbl>    <dbl>    <dbl>
## 1 N               678013   678013   678013   678013
## 2 DF_model            36       36       49       49
## 3 DF_resid        677977   677977   677964   677963
## 4 AIC             286836.  286016.  285809.  285571.
## 5 LogLik         -143382. -142971. -142855. -142735.
## 6 Deviance        217441.  190618.  217101.  217004.
## 7 Pearson_ChiSq  1841084. 1819324. 1771436. 1762631.
## 8 Predicted_Zero  643482.  643084.  644278.  644392.
## 9 Actual_Zero     643953   643953   643953   643953
```

```r
# AIC, LogLik, Pearson, Zero Prediction Accuracy
# Comparing the above metrics, we can see that leading up to the Zero Claims prediction, ZIP and ZINB a
```

```r
#Setting Random Seed for the Project
set.seed(2000)

n <- nrow(french_clean)
train_idx <- sample(seq_len(n), size = 0.7 * n)

french_train <- french_clean[train_idx, ]
french_test  <- french_clean[-train_idx, ]

# Refitting training data set on training data only
zip_train <- zeroinfl(
  ClaimNb ~ VehPower + VehAge + DrivAge + BonusMalus + VehBrand + VehGas + Region +
    offset(log(Exposure)) |
    DrivAge + BonusMalus,
  data = french_train,
  dist = "poisson"
)

# Expected Claim Amount on test data
french_test$mu_hat <- predict(
  zip_train,
  newdata = french_test,
```

```r
  type = "response"
)

# Test-set log-likelihood
test_loglik_zip <- sum(
  dpois(
    french_test$ClaimNb,
    lambda = french_test$mu_hat,
    log = TRUE
  )
)

test_loglik_zip
```

```
## [1] -43243.26
```

```r
# Actual zeros in test set
actual_zeros_test <- sum(french_test$ClaimNb == 0)

# Predicted probability of zero
prob_zero_test <- predict(
  zip_train,
  newdata = french_test,
  type = "prob"
)[, "0"]

predicted_zeros_test <- sum(prob_zero_test)

actual_zeros_test
```

```
## [1] 193147
```

```r
predicted_zeros_test
```

```
## [1] 193303
```

```r
actual_zeros_test - predicted_zeros_test
```

```
## [1] -156.0088
```

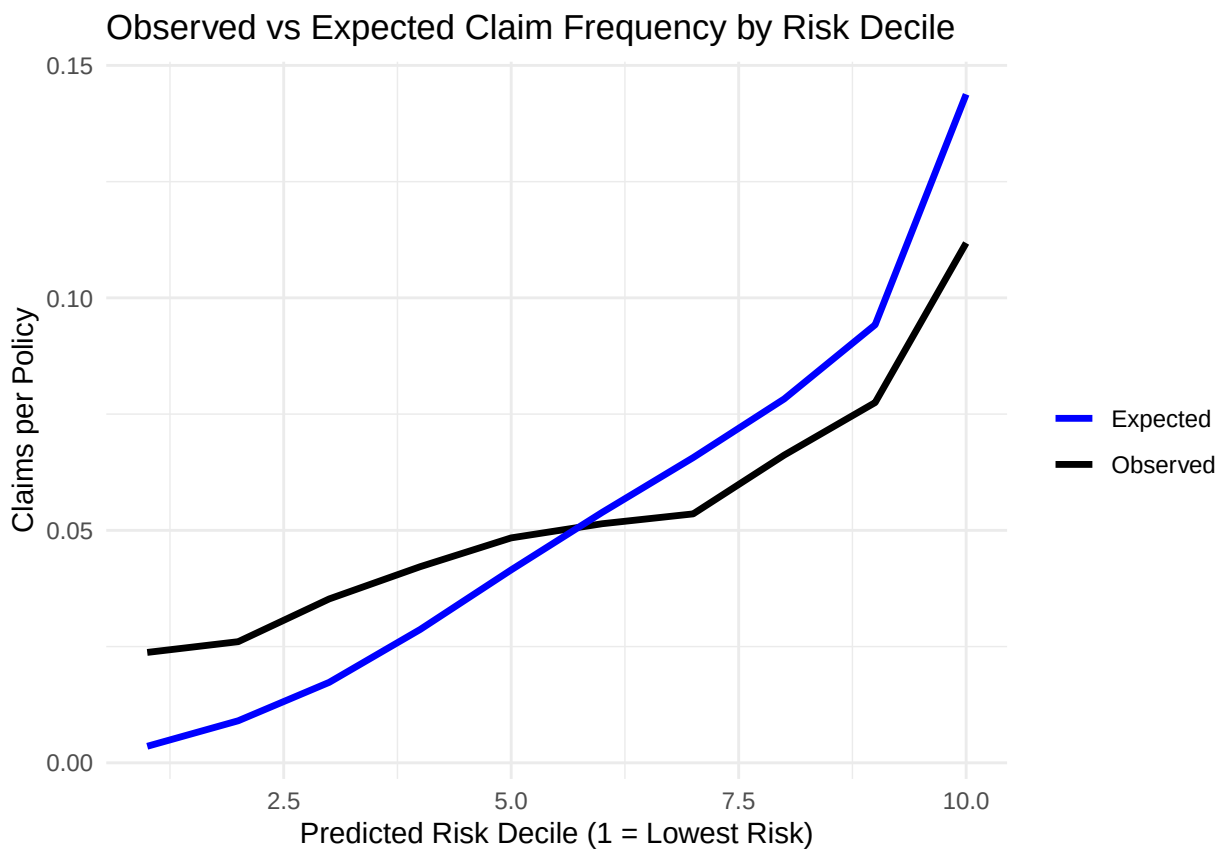
```r
# RMSE
rmse_zip <- sqrt(mean((french_test$ClaimNb - french_test$mu_hat)^2))

# MAE
mae_zip <- mean(abs(french_test$ClaimNb - french_test$mu_hat))

rmse_zip
```

```
## [1] 0.2408242
```

```r
mae_zip
```

```
## [1] 0.099493
```

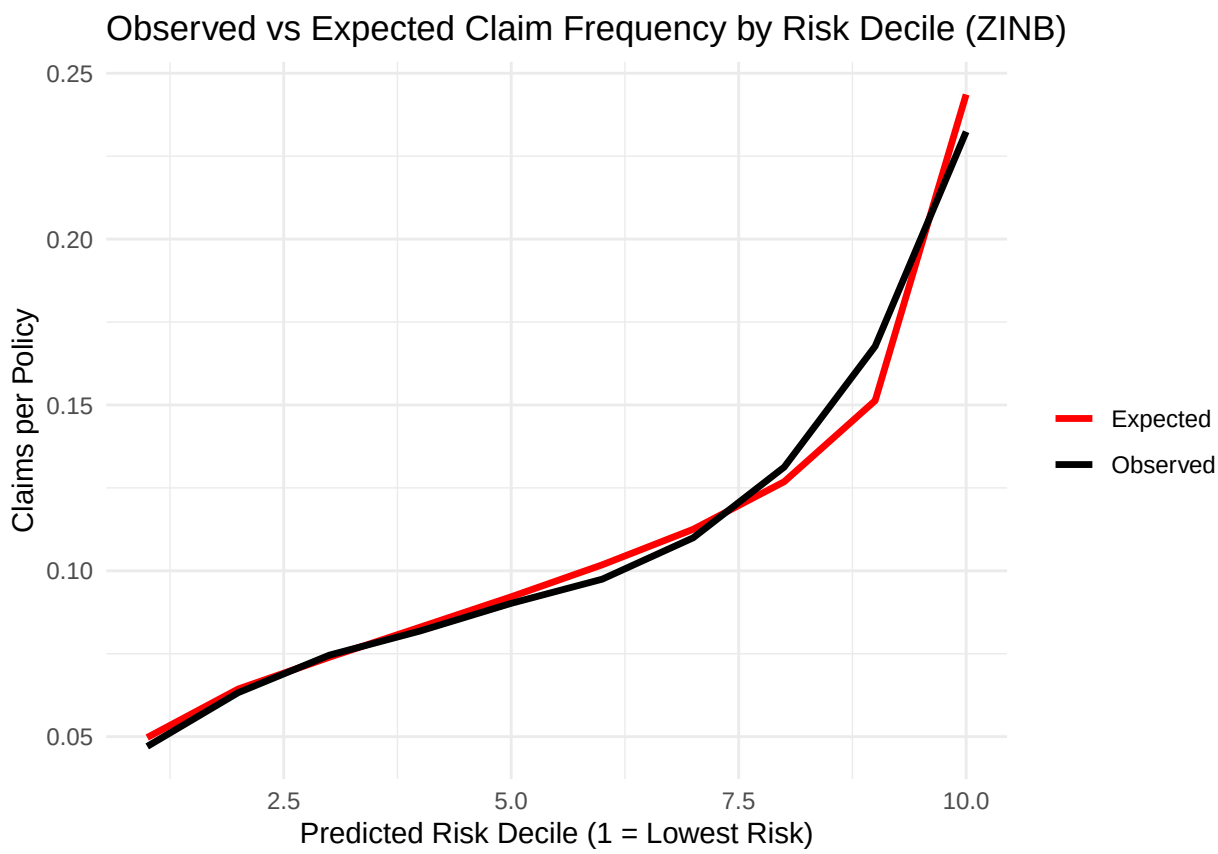
```r
french_test <- french_test %>%
  mutate(
    decile = ntile(mu_hat, 10)
  )

decile_summary <- french_test %>%
  group_by(decile) %>%
  summarise(
    policies = n(),
    observed_claims = sum(ClaimNb),
    expected_claims = sum(mu_hat),
    obs_freq = observed_claims / policies,
    exp_freq = expected_claims / policies
  )

ggplot(decile_summary, aes(x = decile)) +
  geom_line(aes(y = obs_freq, color = "Observed"), linewidth = 1.2) +
  geom_line(aes(y = exp_freq, color = "Expected"), linewidth = 1.2) +
  scale_color_manual(values = c("Observed" = "black", "Expected" = "blue")) +
  labs(
    title = "Observed vs Expected Claim Frequency by Risk Decile",
    x = "Predicted Risk Decile (1 = Lowest Risk)",
    y = "Claims per Policy",
    color = ""
  ) +
  theme_minimal()
```




```
# Main things to look at in the graph - Lines close together, upward trend, not many crossings, no U sh
# Single crossing, 1-5 predicted risk decile observed > expected, 6-10 decile expected > observed
#### NEEDS MAJOR COMMENTS ABOUT KEEPING THE MODEL AS IS
```

```
french_test$mu_hat_zinb <- predict(
  french_zinb,
  newdata = french_test,
  type = "response"
)

french_test <- french_test %>%
  mutate(
    freq_hat_zinb = mu_hat_zinb / Exposure,
    risk_decile_zinb = ntile(freq_hat_zinb, 10)
  )

zinb_decile_summary <- french_test %>%
  group_by(risk_decile_zinb) %>%
  summarise(
    Observed = sum(ClaimNb) / sum(Exposure),
    Expected = sum(mu_hat_zinb) / sum(Exposure),
    .groups = "drop"
  )

ggplot(zinb_decile_summary, aes(x = risk_decile_zinb)) +
  geom_line(aes(y = Expected, color = "Expected"), linewidth = 1.2) +
```

```r
  geom_line(aes(y = Observed, color = "Observed"), linewidth = 1.2) +
  scale_color_manual(values = c("Expected" = "red", "Observed" = "black")) +
  labs(
    title = "Observed vs Expected Claim Frequency by Risk Decile (ZINB)",
    x = "Predicted Risk Decile (1 = Lowest Risk)",
    y = "Claims per Policy",
    color = ""
  ) +
  theme_minimal()
```




```r
# This model ZINB is MUCH better than the ZIP. We can conclude that the ZINB model is better than the Z
```

```r
french_claims <- read_csv("freMTPL2sev.csv")
```

```
## Rows: 26639 Columns: 2
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (2): IDpol, ClaimAmount
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
summary(french_claims)
```

```
##      IDpol          ClaimAmount
##  Min.   :    139   Min.   :      1.0
##  1st Qu.:1087643   1st Qu.:    686.8
##  Median :2137413   Median :   1172.0
##  Mean   :2279864   Mean   :   2278.5
##  3rd Qu.:3180162   3rd Qu.:   1228.1
##  Max.   :6113971   Max.   :4075400.6
```

```r
severity_data <- french_claims %>%
  filter(ClaimAmount > 0)

severity_model_data <- severity_data %>%
  inner_join(
    french_clean,
    by = "IDpol"
  )

# Getting rid of ClaimNb & Exposure

table(severity_model_data$IDpol) |> summary()
```

```
## Number of cases in table: 26444
## Number of factors: 1
```

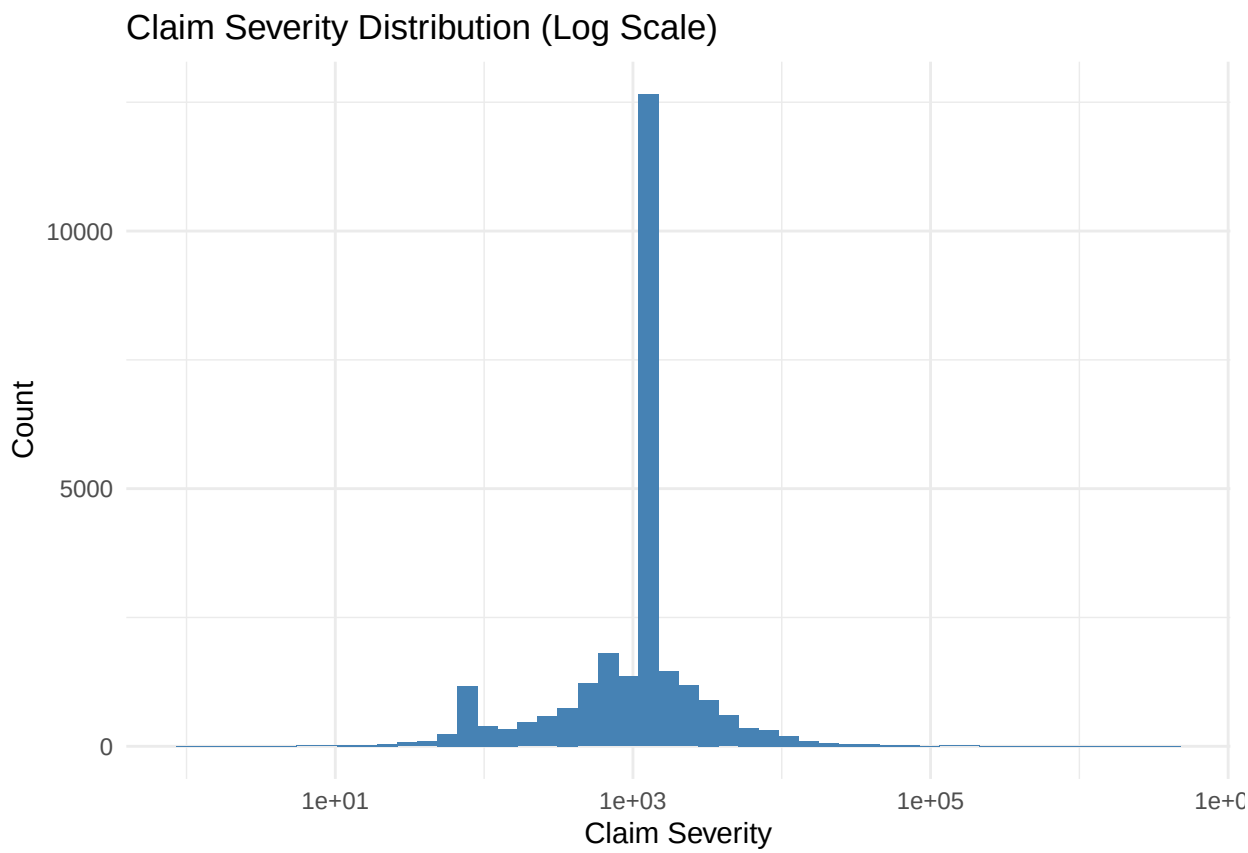
```r
nrow(severity_model_data)
```

```
## [1] 26444
```

```r
summary(severity_model_data$ClaimAmount)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       1     686    1172    2266    1212 4075401
```

```r
ggplot(french_claims, aes(x = ClaimAmount)) +
  geom_histogram(bins = 50, fill = "steelblue") +
  scale_x_log10() +
  theme_minimal() +
  labs(
    title = "Claim Severity Distribution (Log Scale)",
    x = "Claim Severity",
    y = "Count"
  )
```

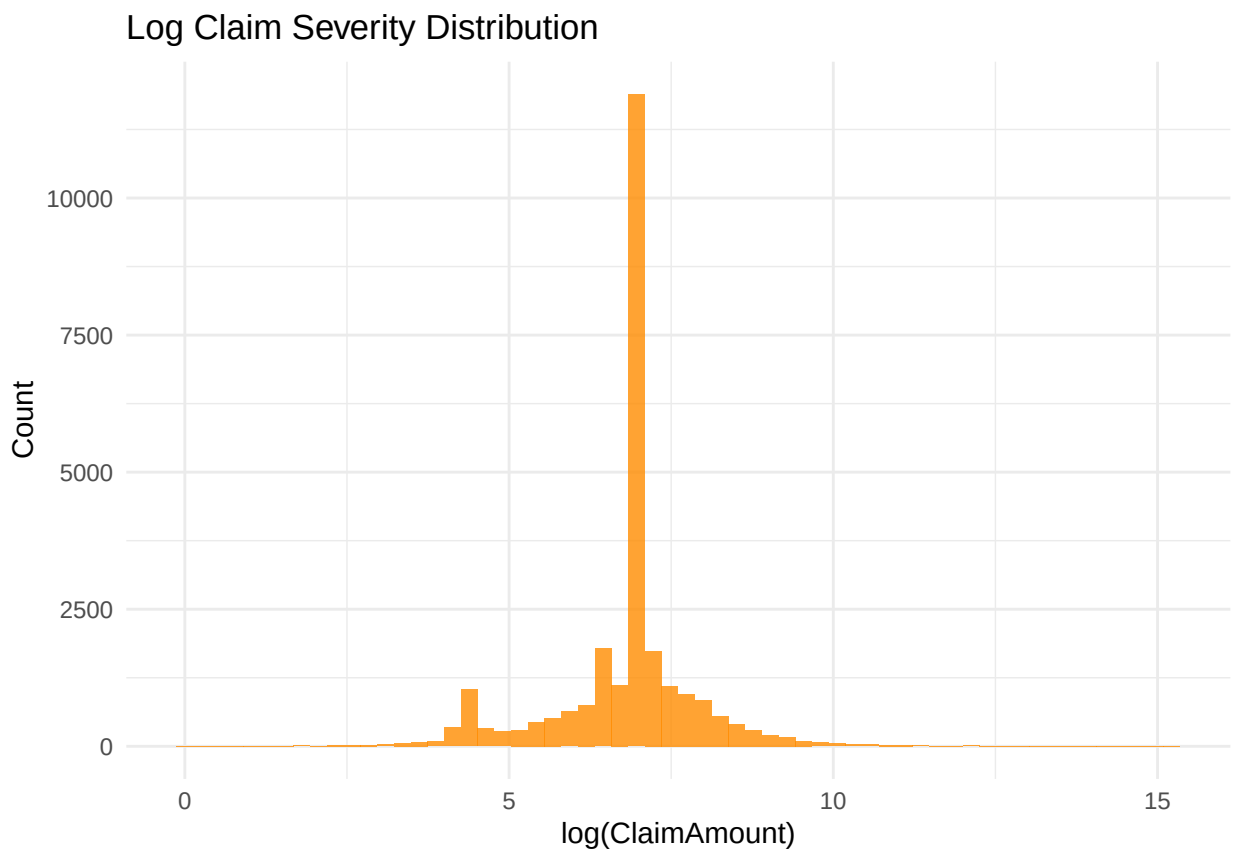



```r
# Using log() to help see the trend of the data better

# Gamma GLM
gamma_model <- glm(
  ClaimAmount ~ VehPower + VehAge + DrivAge + BonusMalus + Density + Region,
  family = Gamma(link = "log"),
  data = severity_model_data
)
summary(gamma_model)
```

```
##
## Call:
## glm(formula = ClaimAmount ~ VehPower + VehAge + DrivAge + BonusMalus +
##     Density + Region, family = Gamma(link = "log"), data = severity_model_data)
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       6.833e+00  7.670e-01   8.909  < 2e-16 ***
## VehPower5                        -2.251e-01  1.446e-01  -1.557  0.11956
## VehPower6                         6.320e-03  1.384e-01   0.046  0.96359
## VehPower7                        -1.465e-02  1.417e-01  -0.103  0.91766
## VehPower8                        -1.546e-01  2.004e-01  -0.772  0.44041
## VehPower9                         5.678e-01  2.237e-01   2.538  0.01114 *
## VehPower10                        7.134e-02  2.240e-01   0.318  0.75014
## VehPower11                       -9.969e-02  2.787e-01  -0.358  0.72054
```

```
## VehPower12                        1.584e-01  4.122e-01   0.384  0.70070
## VehPower13                        9.086e-02  5.897e-01   0.154  0.87755
## VehPower14                        3.436e-01  6.887e-01   0.499  0.61785
## VehPower15                       -3.347e-01  7.067e-01  -0.474  0.63580
## VehAge                            4.248e-04  8.397e-03   0.051  0.95966
## DrivAge                          -1.990e-03  3.246e-03  -0.613  0.53997
## BonusMalus                        7.090e-03  2.395e-03   2.961  0.00307 **
## Density                          -4.009e-06  1.236e-05  -0.324  0.74567
## RegionAquitaine                   3.626e-01  7.428e-01   0.488  0.62545
## RegionAuvergne                    3.408e-02  9.162e-01   0.037  0.97033
## RegionBasse-Normandie             4.827e-01  7.817e-01   0.618  0.53686
## RegionBourgogne                   3.220e-01  8.019e-01   0.401  0.68807
## RegionBretagne                    4.954e-01  7.299e-01   0.679  0.49734
## RegionCentre                      6.543e-01  7.178e-01   0.912  0.36203
## RegionChampagne-Ardenne           1.624e+00  1.055e+00   1.538  0.12396
## RegionCorse                       9.091e-01  9.294e-01   0.978  0.32802
## RegionFranche-Comte               4.275e-01  1.318e+00   0.324  0.74560
## RegionHaute-Normandie             1.887e-01  8.484e-01   0.222  0.82399
## RegionIle-de-France               3.111e-01  7.302e-01   0.426  0.67006
## RegionLanguedoc-Roussillon        3.982e-01  7.427e-01   0.536  0.59186
## RegionLimousin                    1.358e-01  8.631e-01   0.157  0.87498
## RegionMidi-Pyrenees               1.784e-01  7.964e-01   0.224  0.82281
## RegionNord-Pas-de-Calais          4.050e-01  7.352e-01   0.551  0.58170
## RegionPays-de-la-Loire            2.141e-01  7.330e-01   0.292  0.77022
## RegionPicardie                    4.660e-01  8.101e-01   0.575  0.56514
## RegionPoitou-Charentes            2.355e-01  7.525e-01   0.313  0.75429
## RegionProvence-Alpes-Cotes-D'Azur 6.036e-01  7.234e-01   0.834  0.40403
## RegionRhone-Alpes                 5.711e-01  7.202e-01   0.793  0.42778
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 46.63731)
##
##     Null deviance: 46482  on 26443  degrees of freedom
## Residual deviance: 43312  on 26408  degrees of freedom
## AIC: 457061
##
## Number of Fisher Scoring iterations: 12
```

```r
ggplot(severity_model_data, aes(x = log(ClaimAmount))) +
  geom_histogram(bins = 60, fill = "darkorange", alpha = 0.8) +
  theme_minimal() +
  labs(
    title = "Log Claim Severity Distribution",
    x = "log(ClaimAmount)",
    y = "Count"
  )
```




```r
lognorm_model <- lm(
  log(ClaimAmount) ~ VehPower + VehAge + DrivAge + BonusMalus + Density + Region,
  data = severity_model_data)

summary(lognorm_model)
```

```
##
## Call:
## lm(formula = log(ClaimAmount) ~ VehPower + VehAge + DrivAge +
##     BonusMalus + Density + Region, data = severity_model_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.8802 -0.3003  0.1878  0.3756  8.3880
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       6.189e+00  1.262e-01  49.049  < 2e-16 ***
## VehPower5                        -3.110e-02  2.379e-02  -1.307  0.19116
## VehPower6                        -8.816e-03  2.278e-02  -0.387  0.69871
## VehPower7                         4.043e-03  2.332e-02   0.173  0.86235
## VehPower8                         4.615e-02  3.297e-02   1.400  0.16157
## VehPower9                         1.889e-02  3.680e-02   0.513  0.60777
## VehPower10                        8.931e-02  3.686e-02   2.423  0.01538 *
## VehPower11                        2.140e-02  4.585e-02   0.467  0.64073
```

```
## VehPower12                        8.933e-02  6.782e-02   1.317  0.18777
## VehPower13                        9.428e-02  9.702e-02   0.972  0.33116
## VehPower14                       -4.853e-02  1.133e-01  -0.428  0.66845
## VehPower15                       -1.506e-01  1.163e-01  -1.295  0.19531
## VehAge                           -8.134e-03  1.382e-03  -5.887 3.97e-09 ***
## DrivAge                           5.598e-03  5.341e-04  10.481  < 2e-16 ***
## BonusMalus                        5.823e-03  3.940e-04  14.780  < 2e-16 ***
## Density                          -6.185e-06  2.034e-06  -3.041  0.00236 **
## RegionAquitaine                   1.420e-01  1.222e-01   1.162  0.24538
## RegionAuvergne                    9.147e-02  1.507e-01   0.607  0.54399
## RegionBasse-Normandie             4.637e-02  1.286e-01   0.361  0.71839
## RegionBourgogne                   1.282e-01  1.319e-01   0.972  0.33130
## RegionBretagne                    1.147e-01  1.201e-01   0.955  0.33958
## RegionCentre                      4.217e-02  1.181e-01   0.357  0.72101
## RegionChampagne-Ardenne           1.672e-01  1.736e-01   0.963  0.33572
## RegionCorse                       4.125e-01  1.529e-01   2.698  0.00699 **
## RegionFranche-Comte               3.486e-01  2.168e-01   1.608  0.10782
## RegionHaute-Normandie             9.380e-02  1.396e-01   0.672  0.50156
## RegionIle-de-France               1.379e-01  1.201e-01   1.148  0.25110
## RegionLanguedoc-Roussillon        1.713e-01  1.222e-01   1.402  0.16098
## RegionLimousin                    7.148e-02  1.420e-01   0.503  0.61468
## RegionMidi-Pyrenees               9.110e-02  1.310e-01   0.695  0.48690
## RegionNord-Pas-de-Calais          1.526e-01  1.209e-01   1.262  0.20704
## RegionPays-de-la-Loire            3.213e-02  1.206e-01   0.266  0.78991
## RegionPicardie                    6.865e-02  1.333e-01   0.515  0.60649
## RegionPoitou-Charentes            6.432e-02  1.238e-01   0.520  0.60340
## RegionProvence-Alpes-Cotes-D'Azur 1.732e-01  1.190e-01   1.456  0.14550
## RegionRhone-Alpes                 6.291e-02  1.185e-01   0.531  0.59545
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.124 on 26408 degrees of freedom
## Multiple R-squared:  0.01277,    Adjusted R-squared:  0.01147
## F-statistic: 9.763 on 35 and 26408 DF,  p-value: < 2.2e-16
```

```r
AIC(gamma_model, lognorm_model)
```

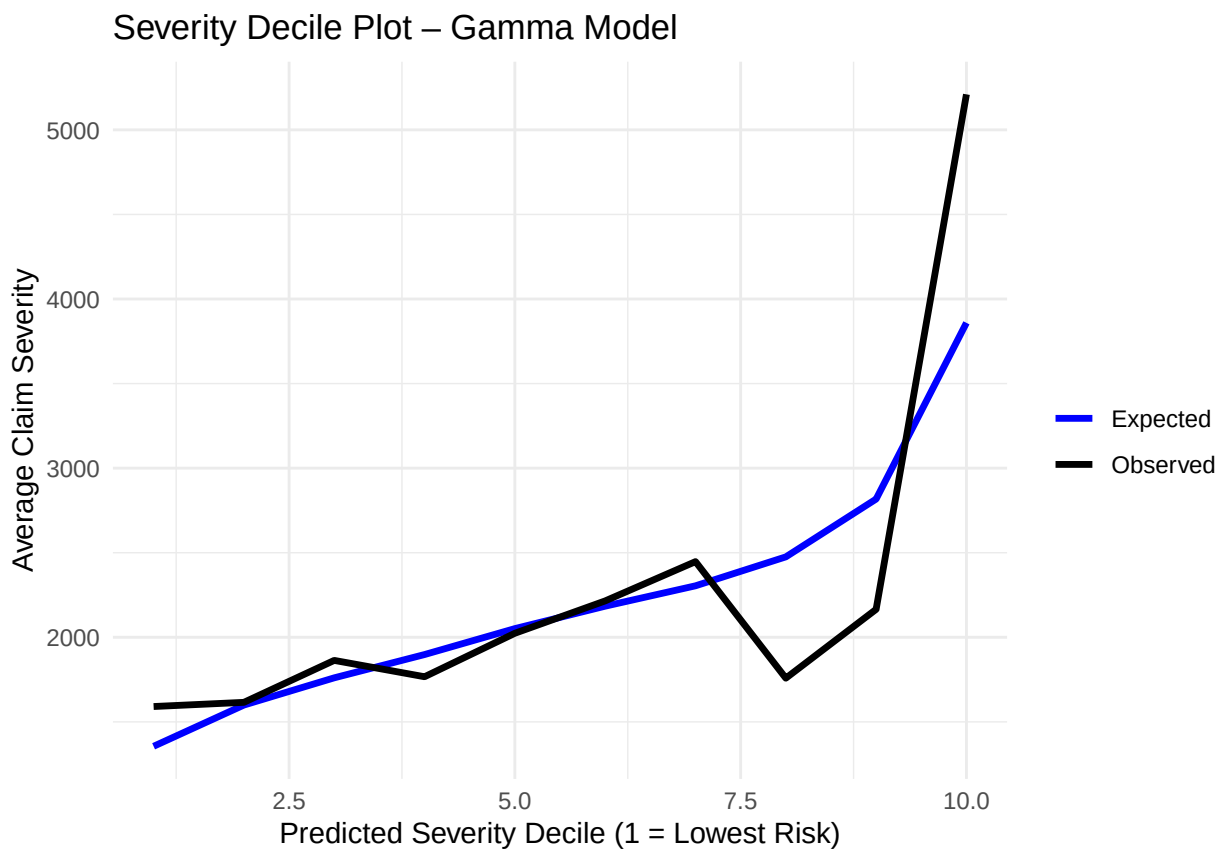
```
##               df       AIC
## gamma_model   37 457061.23
## lognorm_model 37  81242.78
```

```r
# Testing both because the lognormal
# Fits lognormal very well (Low AIC)
# Fits Gamma reasonably well
# Log tested log(ClaimAmount), not comparing apples to apples
```

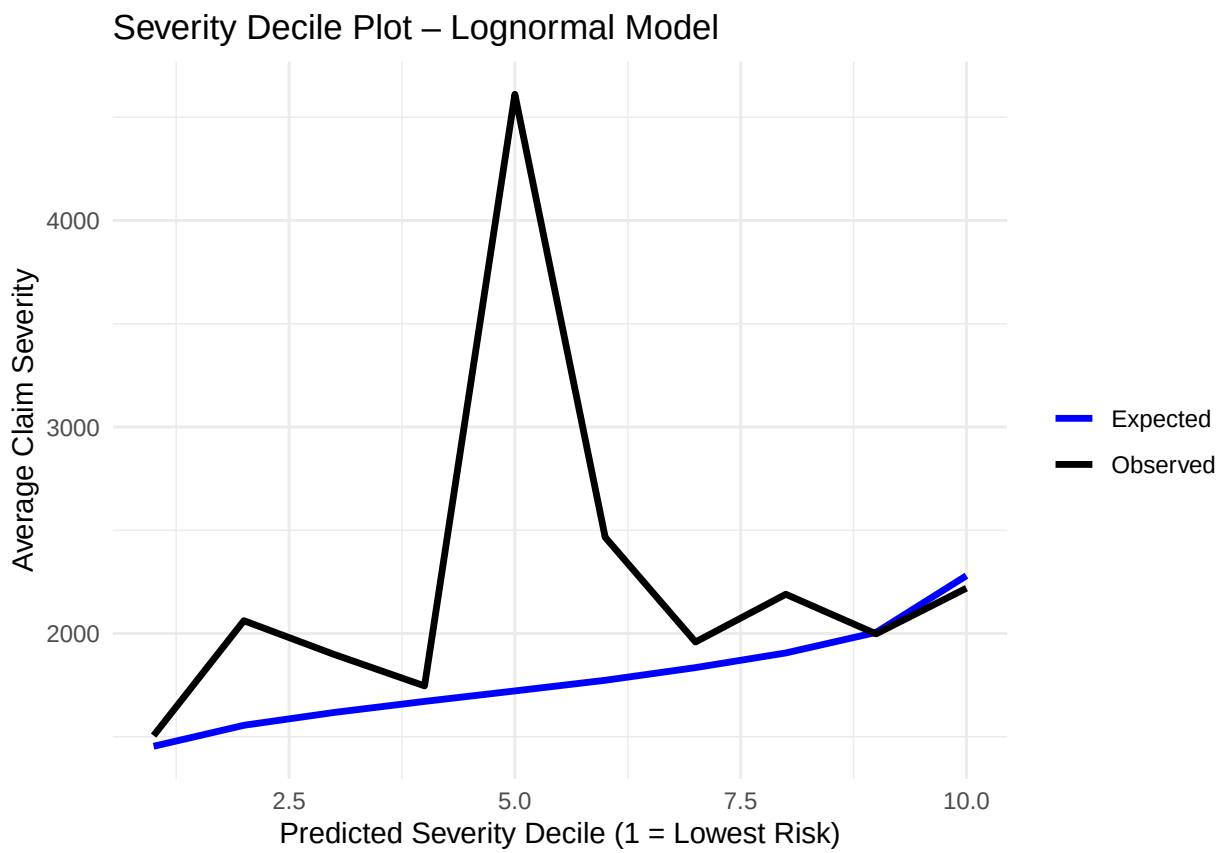
```r
sigma2 <- var(residuals(lognorm_model))
sigma2
```

```
## [1] 1.26064
```

```r
severity_model_data <- severity_model_data %>%
  mutate(
    mu_gamma = predict(gamma_model, type = "response"),
    mu_lognorm = exp(predict(lognorm_model) + 0.5*sigma2)
  )

severity_model_data <- severity_model_data %>%
  mutate(
    gamma_decile    = ntile(mu_gamma, 10),
    lognorm_decile  = ntile(mu_lognorm, 10)
  )

gamma_decile_summary <- severity_model_data %>%
  group_by(gamma_decile) %>%
  summarise(
    Observed = mean(ClaimAmount),
    Expected = mean(mu_gamma),
    .groups = "drop"
  )

lognorm_decile_summary <- severity_model_data %>%
  group_by(lognorm_decile) %>%
  summarise(
    Observed = mean(ClaimAmount),
    Expected = mean(mu_lognorm),
    .groups = "drop"
  )

ggplot(gamma_decile_summary, aes(x = gamma_decile)) +
  geom_line(aes(y = Expected, color = "Expected"), linewidth = 1.2) +
  geom_line(aes(y = Observed, color = "Observed"), linewidth = 1.2) +
  scale_color_manual(values = c("Expected" = "blue", "Observed" = "black")) +
  labs(
    title = "Severity Decile Plot - Gamma Model",
    x = "Predicted Severity Decile (1 = Lowest Risk)",
    y = "Average Claim Severity",
    color = ""
  ) +
  theme_minimal()
```




```r
ggplot(lognorm_decile_summary, aes(x = lognorm_decile)) +
  geom_line(aes(y = Expected, color = "Expected"), linewidth = 1.2) +
  geom_line(aes(y = Observed, color = "Observed"), linewidth = 1.2) +
  scale_color_manual(values = c("Expected" = "blue", "Observed" = "black")) +
  labs(
    title = "Severity Decile Plot - Lognormal Model",
    x = "Predicted Severity Decile (1 = Lowest Risk)",
    y = "Average Claim Severity",
    color = ""
  ) +
  theme_minimal()
```




```r
summary(severity_model_data$ClaimAmount)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       1     686    1172    2266    1212 4075401
```

```r
quantile(severity_model_data$ClaimAmount, probs = c(0.9, 0.95, 0.99, 0.999))
```

```
##        90%        95%        99%      99.9%
##   2767.604   4765.093  16451.224 152223.241
```

```r
# Rather large tail, more problems with lognormal model
```

```r
french_test <- french_test %>%
  mutate(
    freq_hat = predict(zip_train, newdata = french_test, type = "response"),
    sev_hat  = predict(gamma_model, newdata = french_test, type = "response"),
    pure_premium_hat = freq_hat * sev_hat
  )


severity_by_policy <- severity_model_data %>%
  group_by(IDpol) %>%
  summarise(
    total_claim_amount = sum(ClaimAmount),
```

```r
    .groups = "drop"
  )

french_test <- french_test %>%
  left_join(severity_by_policy, by = "IDpol")


french_test$total_claim_amount[is.na(french_test$total_claim_amount)] <- 0

rmse_pp <- sqrt(mean((french_test$total_claim_amount - french_test$pure_premium_hat)^2))
mae_pp  <- mean(abs(french_test$total_claim_amount - french_test$pure_premium_hat))

rmse_pp
```

```
## [1] 3512.066
```

```r
mae_pp
```

```
## [1] 188.1077
```

```r
final_performance <- tibble(
  Model_Component = c("Frequency (ZIP)", "Severity (Gamma)", "Pure Premium"),

  Metric_1 = c("Test LogLik", "AIC", "RMSE"),
  Value_1  = c(test_loglik_zip, AIC(gamma_model), rmse_pp),

  Metric_2 = c("Zero Error", "Tail Behavior", "MAE"),
  Value_2  = c(actual_zeros_test - predicted_zeros_test, NA, mae_pp)
)

final_performance
```

```
## # A tibble: 3 x 5
##   Model_Component  Metric_1    Value_1 Metric_2      Value_2
##   <chr>            <chr>         <dbl> <chr>           <dbl>
## 1 Frequency (ZIP)  Test LogLik -43243. Zero Error      -156.
## 2 Severity (Gamma) AIC         457061. Tail Behavior     NA
## 3 Pure Premium     RMSE          3512. MAE              188.
```